

Nilendu Ganguli

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EXPERIENCE

CITI | SOFTWARE DEVELOPER 2

Aug 2024 – Current | Pune, India

- Designed and Engineered KYCX , a system for generating **disposition summary and risk category assignment for Member Entities** across multiple Regions using **LLMs paired with workflows tailored to CITI** business units.
- This system leverages both **deterministic chains and multi-agent systems**, to process over **290 different types of document categories** defined under KYC taxonomy. Employing the onboarded Gemini 2.5 Pro models, I have deployed a complete solution that handles document discoverability, **efficient batch workflows for vector embeddings of Member Documents(PDFs) and RAG implementation for faster application of Risk Analysis Rules on these docs**. This system is currently being tested as a part of SIT release and aims to **reduce document processing time by 70% and draw 3X more yearly funding** for my team.
- Delivered multiple **backend microservices, micro-batch processing pipelines** to help in the KYC Migration Effort. As per business requirements, used **Fast API apps paired with Unicorn and Gunicorn Servers to create Restful APIs** to fulfill Technical expectations(caters to a wide variety of ASGI and WSGI apps).
- Impact: Risk of Non-Compliance: Introduced much needed leanness in legacy KYC. Increased throughput by 6X without altering resource utilization. Yielded a **reduction Response time from 230ms to 110 ms**.

CITI | SOFTWARE DEVELOPER

Jul 2022 – Aug 2024 | Pune, India

- Launched **KYC Document Processing and Archival Service** in North America, that eliminated vendor dependency for KYC and laid foundation for implementation of **KYC 2.0 in NAM and Europe** .
- Created **microservices to support CRUD operations** for KYC 1.0 compatibility and **implemented Luigi and Apache NIFI workflows** to enhance KYC 2.0, thereby processing upwards of **100K KYC Documents per day** and resulting in data and service availability for downstream KYC Services.
- Impact: **Completely automated KYC document processing**(mostly PDFs), storage and indexing workflows. **Reducing Client on-boarding time from 14 days to 2 days**.

INTERNSHIPS

CITI | SOFTWARE DEVELOPER

Jun 2021 - Jul 2021 | Pune, India

- Designed and Implemented a **BigData** framework for **Global Data Purge Policy(Europe)** on **KYC Production Data Warehouse**.
- Created **Apache Spark and Kafka Pipeline** for implementation of GDPR policy on Archival data for **CitiBank Member Entity** amounting to uncompressed **size of 43 TeraByte**.
- Deployed and tested the distributed system framework in pre-production environments to ensure **scalability and efficiency** for handling **Big Data velocities in Production** . **Eliminated Risk of non-compliance with GDPR Policies** that may attract **millions of dollars in fines** if violated.

ORIGIN HEALTH | DATA SCIENCE INTERN

Mar 2020 - Aug 2020 | Singapore

- Developed an automated pipeline for backend production for **Image segmentation maps using medical ultrasound images**.
- Techstacks such as WandB and AWS were used for Visualization and Deployment of ML models.
- A novel version of U-Net called **DRUnet** was used to achieve **IoU of 0.74**.

SKILLS

PROGRAMMING

Python • JavaScript
Go • SQL • Java • C++

FRAMEWORKS

FastAPI • Apache Spark • Pandas
Redis • RabbitMQ • Postgres
Apache Kafka • NginX • REST
Uvicorn • Hadoop • PyTorch
Numpy • OracleDB • Sqoop
Kibana • Spring • GraphQL
MongoDB • Elasticsearch

PLATFORMS

Git • Openshift • Cloudera
Kubernetes • AWS • Docker

EDUCATION

NIT TRICHY

BACHELOR OF TECHNOLOGY

Graduation : May 2022

COURSEWORK

Data Structures And Algorithms
Operating Systems
Database Management Systems
Networks And Protocols
BigData Analytics
Machine Learning and Deep Learning
Software Engineering
Industrial Robotics

ACHIEVEMENTS

1st Position at EYRC, International Robotics Competition (AI and Robotics Competition at IIT Bombay).

Successfully planned and implemented **Negative News Tracking and Risk Score Pipeline** for improving KYC Work Item resolution time. Maintained these respective Data Warehouses for 14 Months overseeing **74 TB of Data**.